

Algebra First-Year Exam, August 2017, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. Show that the ring $\mathbb{Z}[\sqrt{-5}]$ is not a principal ideal domain, and give an example of a non-principal ideal in this ring.
2. Let R be a commutative ring with identity. Show that an element $a \in R$ is contained in every maximal ideal if and only if $1 + ab$ is a unit in R for all $b \in R$.
3. Show that if p is an odd prime then the polynomial $\frac{(x+2)^p - 2^p}{x}$ is irreducible in $\mathbb{Q}[X]$.
4. Let K be a field. Show that any K -vector space V has a basis (do not assume that V has finite dimension).
5. Suppose R is a principal ideal domain and M is a finitely generated R -module. For any prime $p \in R$, let M_p be the set of $m \in M$ such that $p^n m = 0$ for some n .
 - (i) Show that M_p is a submodule of M .
 - (ii) Show that if $rM = 0$ for some nonzero $r \in R$, then M is isomorphic to the direct sum of the M_p for all primes p dividing r .
6. Find representatives of all conjugacy classes of elements of order 5 in $GL_2(\mathbb{F}_{19})$. Hint:

$$x^5 - 1 = (x - 1)(x^2 - 4x + 1)(x^2 + 5x + 1)$$

in $\mathbb{F}_{19}[x]$, and the quadratic factors are irreducible.